

1 **Wild bird detection on dairy farms using a training-free multi-stage computer**
2 **vision pipeline**

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ABSTRACT

Wild birds pose persistent biosecurity and productivity challenges on dairy farms, but most operations lack systematic methods to monitor bird activity over time. This study evaluated a sequential, training-free computer vision pipeline for wild bird detection using field-collected video data from a commercial dairy farm. A total of 53,948 motion-filtered frames were analyzed across two monitoring locations over a 30-day period during peak summer bird activity. The pipeline combined (1) region-of-interest filtering, (2) appearance-based candidate detection using adaptive thresholding and Hue, Saturation, and Value (HSV) sky-background classification with cloud rejection, (3) Farneback optical-flow-based motion discrimination, and (4) nearest-neighbor temporal confirmation to reduce transient detections and repeated counting across consecutive frames. Detection performance was evaluated against manual image-based reference counts using frame-level precision, recall, and F1 score. Across strategy-location combinations, precision ranged from 0.71 to 0.92, recall from 0.62 to 0.84, and F1 score from 0.69 to 0.88. Fixed-threshold detection achieved the highest observed precision, whereas the highest recall was observed under fixed-threshold and adaptive-threshold configurations. Motion-based discrimination produced the largest performance improvement, increasing mean F1 score from 0.43 to 0.69 by reducing false positives while maintaining sensitivity to sustained bird movement. Temporal confirmation further improved detection stability and reduced short-term fluctuations in detected counts. Detection performance varied by monitoring location, indicating that background complexity and environmental conditions influenced pipeline performance. These findings support sequential, training-free computer vision pipelines as practical tools for estimating relative bird activity and bird pressure in agricultural settings.

37 Keywords: Bird detection; computer vision; agricultural monitoring; biosecurity

38 INTRODUCTION

39 Wild birds are a persistent problem on dairy farms, causing feed losses and significant
40 biosecurity risks. Species such as European starlings (*Sturnus vulgaris*), pigeons (*Columba livia*),
41 and house sparrows (*Passer domesticus*) have been linked to feed contamination and pathogen
42 spread, including *Salmonella* detected in up to 12% and *E. coli* in up to 80% of bird fecal
43 samples (Adams-Progar et al., 2020). Birds consume between 2.8% and 4.9% of a farm's daily
44 feed supply, representing a meaningful economic loss at the herd level, and selective feeding
45 behavior can substantially alter the nutrient composition of exposed feed, with reported
46 reductions of up to 83% in key components such as dry matter, protein, and starch under heavy
47 bird pressure (Depenbusch et al., 2011; Shwiff et al., 2012). Wild birds are also recognized
48 reservoirs for highly pathogenic avian influenza (HPAI) viruses, and recent North American
49 outbreaks have underscored their role in virus dispersal across agricultural landscapes along
50 migratory routes, with recent HPAI H5N1 events in dairy cattle highlighting the relevance of
51 farm-level surveillance and biosecurity preparedness (Elbers and Gonzales, 2020; Rodriguez et
52 al., 2025, 2024b). Understanding when and where bird activity occurs on farms is therefore
53 directly relevant to both productivity management and biosecurity planning.

54 Despite these recognized impacts, most dairy operations lack systematic and automated methods
55 to identify and track wild bird presence over time. Producer-based estimates of bird pressure
56 often recorded using broad categories such as low, medium, or high, can support management
57 decisions (Shwiff et al., 2012). However, these estimates are inherently subjective, require
58 human observation, and may miss bird activity during early morning hours, periods of low
59 human presence, or areas that are monitored less frequently. These limitations restrict the ability

60 to characterize temporal patterns of bird activity and cumulative exposure across different farm
61 locations. Addressing these limitations requires automated monitoring systems capable of
62 capturing wild bird activity consistently across time and farm locations. Rather than relying only
63 on isolated image-level counts, repeated detections of wild bird activity can be summarized as
64 bird pressure, a broader measure of potential exposure that incorporates the frequency, duration,
65 and location of bird activity near cattle, feed, or water sources.

66 Computer vision has emerged as a feasible approach for automated bird detection and counting
67 (Ma et al., 2024). Prior studies have demonstrated the feasibility of image- and video-based
68 detection using deep learning models, including YOLO-based architectures (Ma et al., 2024), as
69 well as applications in bird counting and behavior recognition (Akçay et al., 2020). However,
70 these methods encounter persistent challenges related to background complexity, lighting
71 variability, small target size, and transient non-avian motion. These challenges are further
72 exacerbated in agricultural environments by cloud drift, structural obstructions, and moving
73 livestock within the camera field of view (Weinstein, 2018; Yang et al., 2024). In addition, deep
74 learning approaches typically require GPU-enabled hardware and large labeled datasets, and
75 their performance is often sensitive to domain shift across farms and camera configurations
76 (Weinstein, 2018; Yang et al., 2024). These constraints can limit scalability and consistent
77 performance in practical farm settings. As a result, lightweight, training-free pipelines that can be
78 deployed on consumer-grade hardware without specialized expertise represent an important but
79 underexplored alternative. This study evaluates complementary computer vision design strategies
80 applied sequentially as a practical detection pipeline using field-collected videos from a
81 commercial dairy facility.

82 The objectives of this study were to (1) evaluate the performance of a multi-stage, training-free
83 computer vision pipeline for detecting wild bird activity under dairy farm field conditions; (2)
84 quantify the contribution of motion-based discrimination and temporal confirmation to detection
85 performance and output stability; and (3) assess how background complexity and environmental
86 conditions influence the use of automated detections as relative indicators of bird pressure over
87 time.

88 **METHODS & MATERIALS**

89 ***Study Location and Design***

90 This study was conducted at the Michigan State University Dairy Teaching and Research Center,
91 a commercial-scale dairy facility housing approximately 350 lactating Holstein cows and 50
92 calves. The facility includes feeding alleys, covered feed storage structures, and outdoor feed
93 storage areas that are routinely exposed to bird activity. In consultation with facility staff, two
94 monitoring locations were selected to represent areas with frequent bird presence and potential
95 contact with cattle, feed, or farm infrastructure. Video data were collected over a 30-day period
96 during July and August 2025 in those same locations.

97 ***Camera Equipment and Setup***

98 Video data were collected using a GoPro Hero 13 consumer-grade camera. The camera was
99 mounted on an adjustable tripod and positioned approximately 30 cm above ground level, angled
100 upward to capture sky-oriented flight paths. Recordings were initiated manually each day at 8:00
101 am and consisted of consecutive 30 minutes segments collected over a three-hour period,
102 concluding at 11:00 am.

Videos were recorded at a resolution of 1920×1080 pixels (16:9 aspect ratio) using a wide-angle lens at 30 frames per second. To mitigate overheating during direct sunlight exposure, batteries were replaced every 30 minutes.

Video Data Processing

Raw video files were transferred daily to a secure external hard drive and reviewed to confirm camera stability, field of view, and absence of obstructions. To reduce computational burden associated with full-frame processing, a preliminary motion-based frame extraction step was applied. A Python-based motion detection script was used to identify frames containing pixel-level movement, thereby restricting downstream analysis to frames likely to contain bird activity. Frame processing rate (frame skip) was varied across implementations to balance detection accuracy, computational efficiency, and avoidance of repeated detections of the same bird across consecutive frames. Lower frame skip values increase temporal resolution but may result in duplicate detections, whereas higher frame skip values reduce redundancy at the cost of potentially missing brief events. Motion-filtered video segments were stored in AVI format and used as input for the evaluation of detection strategies described below. This pre-filtering step was applied uniformly across all recordings and was not intended to classify or identify bird activity, but solely to reduce data volume for subsequent analysis.

Overview of Evaluated Detection Strategies

Three pipeline configurations were developed to evaluate different detection strategies: (i) sky-background filtering, (ii) fixed-threshold, and (iii) adaptive thresholding.

(i) *Sky-background filtering*: Implements fixed region-of-interest (ROI) boundaries with enhanced sky-background classification in HSV color space, optical flow-based motion

126 discrimination, and cloud rejection using texture analysis. Frames were subsampled by
127 processing every second frame to reduce repeated detections.

128 (ii) *Fixed-threshold (frame-based)*: Implements a fixed frame coverage approach (60% of frame
129 height), combined with background subtraction using a Gaussian mixture model (MOG2) and
130 simplified motion detection via frame differencing. All frames were processed without
131 subsampling, with image resizing applied to improve computational efficiency.

132 (iii) *Adaptive threshold*: Implements dynamic ROI adjustment using Sobel-based horizon
133 detection to adapt the analysis region to scene changes (e.g., vehicles entering the frame). This
134 approach combines sky-background filtering with optical flow analysis. All frames were
135 processed without subsampling, with periodic ROI updates applied during processing. Adaptive-
136 threshold processing was applied without subsampling because dynamic ROI updates and
137 optical-flow estimation require temporally dense input to maintain stable motion estimates
138 between consecutive frames.

139 Each pipeline represents a distinct detection configuration designed to address specific sources of
140 error under field conditions. To achieve this, complementary computer vision strategies were
141 applied sequentially rather than as competing standalone approaches. The evaluated strategies
142 operate at different stages of the detection process and target specific classes of error
143 encountered during deployment. Pipelines were implemented as progressively refined
144 configurations, reflecting practical system development rather than isolated component ablations
145 (Fig. 1). Accordingly, each pipeline included four core components, (a) region-of-interest (ROI)
146 filtering to restrict analysis to image regions where birds in flight are expected; (b) appearance-
147 based filtering, combining adaptive thresholding, contour-based candidate detection, and sky-
148 background identification to generate detection candidates; (c) motion-based discrimination

149 using optical flow analysis to distinguish bird movement from background motion; and (d)
150 temporal confirmation, employing multi-frame tracking and persistence criteria to suppress
151 transient detections and prevent double-counting.

152 ***Pipeline Design and Implementation***

153 ***Region of Interest Processing.*** A region-of-interest (ROI) filtering system was implemented to
154 restrict analysis to relevant portions of each video frame and reduce interference from non-bird
155 elements. Two ROI strategies were used: (i) fixed ROI, applied in the sky-background filtering
156 and fixed-threshold pipelines, and (ii) dynamic ROI, applied in the adaptive threshold pipeline.

157 The fixed ROI strategy used predefined vertical boundaries expressed as percentages of total
158 frame height. In the sky-background filtering pipeline, the ROI was restricted to the upper half
159 of the frame. In contrast, the fixed-threshold (frame-based) pipeline used a broader coverage
160 encompassing most of the frame. These configurations were selected based on observed bird
161 flight patterns at each monitoring location, where most activity occurred in the upper portion of
162 the frame.

163 The dynamic ROI strategy used in the adaptive threshold pipeline adjusted the lower boundary of
164 the analysis region based on scene characteristics to maintain focus on sky regions while
165 excluding transient ground-level objects. The sky-ground boundary was estimated using
166 gradient-based edge detection. A Sobel operator was applied to a blurred frame to detect
167 horizontal edges, and a vertical projection of edge magnitude was used to identify regions of
168 high edge density. The algorithm then searched upward from this peak until edge density fell
169 below a predefined threshold, indicating the sky-ground boundary.

170 To improve stability, ROI boundaries were temporally smoothed using a weighted average of
171 current and newly computed boundaries:

$$172 \quad \textit{smoothed_roi} = \textit{current_roi} \times 0.5 + \textit{new_roi} \times 0.5$$

173 ROI boundaries were recalculated at regular intervals and constrained within predefined limits to
174 prevent extreme values.

175 ***Appearance-Based Filtering.*** In outdoor video collected under variable natural lighting, a fixed
176 global threshold applied uniformly across an entire frame will fail to correctly segment targets
177 when illumination varies spatially, for instance, when part of the sky is directly lit while another
178 region is overcast or shaded. This has been identified in previous studies as a shortcoming of
179 global thresholding approaches (Rosebrock, n.d.). To address this, adaptive thresholding is used,
180 selecting the optimal binarization threshold for each pixel neighbourhood rather than a single
181 value per frame. Specifically, a Gaussian-weighted kernel is applied to determine the threshold
182 for each pixel neighbourhood, enabling robust segmentation under spatially varying illumination
183 conditions (Rosebrock, n.d.). Before thresholding, the image is resized for computational
184 efficiency and smoothed using a Gaussian kernel to reduce sensor noise. Parameter values varied
185 by implementation (Table 1).

186 Contours are extracted from the binarized frame using the border-following algorithm of Suzuki
187 and Abe (Suzuki & Be, 1985), which underpins OpenCV's findContours function. The algorithm
188 traverses connected boundaries of binary regions topologically, extracting candidate blobs
189 without requiring spatial convolution. Contour area and shape thresholds were empirically
190 derived from annotated frames to capture the range of bird silhouette sizes and orientations
191 observed across species and flight postures, while excluding noise blobs, structural edges, and

192 vegetation features. Accurate identification of candidate objects against sky backgrounds
193 requires a colour representation that is robust to illumination variation. HSV colour space was
194 used for sky-background classification. Candidate pixels were evaluated using predefined hue,
195 saturation, and value ranges corresponding to sky conditions observed during field deployment.
196 These ranges were used to classify candidate regions as sky-associated or non-sky-associated
197 before subsequent cloud-rejection and motion-discrimination steps.

198 The system used a multi-condition sky classification scheme, with separate HSV acceptance
199 ranges defined for four atmospheric states encountered during field deployment: clear blue sky,
200 white and light grey sky, light blue and cyan sky, and grey or overcast sky. Candidate pixels
201 were grouped with neighbouring pixels and retained only if they satisfied at least one of the sky-
202 condition criteria. Low-saturation conditions such as overcast grey relied more heavily on the
203 Value channel for classification, while high-saturation conditions such as clear blue sky were
204 more strongly discriminated by Hue.

205 Cloud boundaries are a major source of false positives in bird detection within sky regions, as
206 cloud edges can produce contours that overlap with bird identification criteria. To address this, a
207 multi-feature rejection stage simultaneously examines texture, edge sharpness, and colour
208 uniformity within each candidate region. Local texture is measured as the average standard
209 deviation of pixel intensity within a small neighbourhood, values below a defined threshold
210 indicate cloud-like smoothness. Edge density is calculated as the ratio of edge pixels to total
211 pixels, with low values indicating the soft boundaries typical of cloud surfaces. Colour
212 uniformity is assessed as the standard deviation of HSV saturation values within the region, with
213 low values indicating the uniform colour characteristic of cloud formations. A candidate region
214 is classified as cloud-like and rejected when it exhibits low texture variation, low edge density,

215 and high color uniformity, traits of diffuse cloud surfaces rather than the sharper, more varied
216 outlines of birds in flight.

217 ***Motion-Based Discrimination Using Optical Flow.*** Birds in flight exhibit motion patterns that
218 differ markedly from the slow, uniform drift of clouds. They move faster, follow non-linear
219 paths, and change direction frequently due to wing beats and course corrections. To capture these
220 differences, the system uses Farneback dense optical flow analysis to compute per-pixel motion
221 vectors between consecutive frames. Farneback flow was chosen over sparse optical flow
222 methods because it covers the entire frame rather than relying on pre-selected feature points,
223 which may not perform well for small, fast-moving targets.

224 For each candidate region passing appearance-based filtering, three motion parameters are
225 assessed. Flow magnitude indicates movement speed and is accepted within a defined range.
226 Consistency of flow direction is measured as the ratio of mean flow vector magnitude to the
227 average of individual vector magnitudes. Birds often change direction during flight, resulting in
228 lower consistency values; a defined directional change threshold helps distinguish purposeful
229 flight from steady atmospheric drift. Motion detection parameters vary by implementation
230 strategy (Table 2).

231 The sky-background filtering and adaptive threshold implementations use Farneback optical
232 flow for detailed motion analysis, while fixed-threshold uses computationally lighter frame
233 differencing suited to faster processing. Candidates that do not meet these criteria, particularly
234 those with high directional consistency and low flow magnitude typical of cloud drift, are
235 excluded from further analysis. All thresholds were established through visual review of motion-
236 filtered outputs from video samples collected at each monitoring site.

237 ***Temporal Confirmation Through Flight Pattern Recognition and Tracking.*** Detecting bird
238 candidates frame by frame, even after appearance and motion filtering, can still produce transient
239 false positives when non-bird features temporarily satisfy detection criteria in isolated frames.
240 Temporal confirmation addresses this by requiring that a detected object appear persistently
241 across multiple consecutive frames along a consistent path before being classified as a confirmed
242 bird. The implementation uses a nearest-neighbour matching algorithm to associate candidate
243 detections across frames, permitting a defined maximum spatial displacement between frames.
244 This straightforward association approach reduces computational load on standard hardware
245 where GPU-accelerated predictive tracking is unavailable.
246 An object is confirmed as a flying bird when it meets persistence and movement criteria, which
247 vary by implementation (Table 3).

248 ***Reference Annotation and Manual Bird Counts***

249 To provide an independent reference for evaluating detection outputs, a subset of video frames
250 was manually reviewed by two researchers who were not involved in the development or
251 implementation of the computer vision pipeline. Reviewers worked independently and recorded
252 the number of visible birds in each reviewed frame based on visual inspection of continuous
253 video playback. For each reviewed frame, the average of the two reviewer counts was used as the
254 manual reference count for performance evaluation. Manual annotations were intended to
255 provide practical field reference counts for comparing detection behavior across pipeline
256 configurations. These reference counts were used to assess agreement with automated outputs,
257 including false positives, missed detections, and temporal consistency across the evaluated
258 detection strategies.

259 ***Performance Metrics and Statistical Summaries***

260 Detection performance was evaluated by comparing automated bird counts with manual image-
261 based reference counts in annotated video frames. Manual review was conducted by two
262 independent researchers who were not involved in the development or implementation of the
263 computer vision pipeline. For each reviewed frame, the number of visible birds was recorded by
264 visual inspection and used as the reference count. Manual annotation was intended to provide a
265 practical field reference for comparing detection behavior across pipeline configurations, rather
266 than an exhaustive census of all bird activity.

267 Performance was summarized using precision, recall, and F1 score. Metrics were calculated at
268 the frame level using a count-based matching approach. For each annotated frame i , the
269 automated bird count (a_i) was compared with the manual reference count (m_i). True positives
270 were defined as $TP_i = \min(a_i, m_i)$, false positives as $FP_i = \max(0, a_i - m_i)$, and false negatives as
271 $FN_i = \max(0, m_i - a_i)$. Precision and recall were calculated across all annotated frames as $\Sigma TP /$
272 $\Sigma(TP + FP)$ and $\Sigma TP / \Sigma(TP + FN)$, respectively. The F1 score was calculated as $2 \times \text{precision} \times$
273 $\text{recall} / (\text{precision} + \text{recall})$.

274 False positives represented automated counts that exceeded the manual reference count in a
275 frame and included non-bird objects, atmospheric artifacts, cloud edges, or moving background
276 elements incorrectly classified as birds. False negatives represented manually observed birds that
277 were not counted by the automated pipeline, including birds missed because of small apparent
278 size, low contrast, distance from the camera, dark or cloudy backgrounds, or partial occlusion.

279 Of the 53,948 motion-filtered frames processed by the pipeline, a subset of annotated frames was
280 used for performance evaluation. The remaining frames were processed by the automated
281 pipeline but were not included in metric calculation because exhaustive manual annotation was
282 infeasible. Performance metrics were summarized descriptively by detection strategy and

283 monitoring location. No formal hypothesis testing was performed because the objective was to
284 compare detection behavior across pipeline configurations rather than estimate population-level
285 parameters or select a single optimized model.

286 Because the applied goal was to characterize bird pressure over time rather than estimate the
287 exact number of unique individual birds, temporal outputs were interpreted as relative indices of
288 bird activity or exposure intensity. Repeated detections across adjacent frames were not
289 considered inherently problematic for this objective, provided that detection behavior was
290 sufficiently stable to preserve temporal and spatial patterns. To reduce excessive inflation from
291 repeated detections, temporal confirmation and maximum concurrent count summaries were
292 used to stabilize detection outputs across recording intervals. For each recording interval, the
293 maximum number of birds detected simultaneously was calculated as a summary of bird
294 pressure, reducing the influence of repeated detections of the same individual across adjacent
295 frames.

296 Detection outputs were also summarized over time to evaluate consistency across recording
297 intervals and to assess the stabilizing effect of motion-based discrimination and temporal
298 confirmation. These temporal summaries were interpreted descriptively as indicators of relative
299 bird activity under field conditions, rather than absolute counts of unique birds visiting the farm.

300 ***Visualization of Detection Outputs***

301 Detection results were visualized during processing to support verification of detection
302 behaviour at each stage of the evaluation pipeline. Detected objects are shown with colour-coded
303 bounding boxes reflecting their classification stage: cyan for initial detections, yellow for sky-
304 region objects, green for those exhibiting bird-like motion, and red for confirmed flying birds.
305 Performance statistics are displayed in real time, including frame count, unique birds detected,

306 and counts at each pipeline stage, enabling immediate evaluation and parameter adjustment for
307 specific environmental conditions.

308 ***Automated Output Logging and Data Export***

309 Detection results were automatically recorded in both human-readable text and CSV formats.
310 Logged metrics included the total number of unique flying birds detected, the maximum number
311 of concurrently visible birds, processing efficiency measured in frames per second, and
312 timestamped duration data for each recording interval. CSV outputs supported subsequent
313 statistical analysis and integration with other ecological monitoring processes. For batch
314 processing, summarised statistics were produced across multiple video files to enable efficient
315 analysis of extended monitoring periods.

316 **RESULTS**

317 ***Descriptive Summary***

318 A total of 53,948 motion-filtered image frames were processed across multiple video segments
319 collected at two monitoring locations (A and B). Image-based manual bird counts were provided
320 as a practical reference to evaluate detection behavior across alternative appearance-based and
321 progressively refined detection strategies. Manual annotations were not intended to represent an
322 exhaustive census, but rather to support comparative assessment of false detections, missed
323 detections, and temporal consistency across recordings.

324 ***Performance of Appearance-Based Detection Strategies***

325 Appearance-based detection performance varied across the three evaluated strategies (Table 4).
326 All strategies were evaluated following application of ROI filtering that restricted analysis to
327 sky-dominated regions, thereby reducing ground-related false detections prior to quantitative
328 assessment.

329 Sky-background filtering achieved precision values of 0.75 at Location A and 0.83 at Location
330 B, with corresponding recall values of 0.81 and 0.62, indicating a trade-off between reduced
331 false detections and sensitivity. However, this came at the cost of reduced recall in some
332 conditions, reflecting missed detections under variable lighting and background conditions.

333 Adaptive thresholding achieved precision values of 0.78 at Location A and 0.71 at Location B,
334 with recall values of 0.84 and 0.67, respectively. This increase in sensitivity was frequently
335 accompanied by increased false positives associated with atmospheric features, resulting in
336 greater variability in precision across recordings.

337 Fixed thresholding achieved precision values of 0.92 at Location A and 0.89 at Location B, with
338 corresponding recall values of 0.84 and 0.67. While this strategy achieved relatively balanced
339 performance in some recordings (F1 scores up to 0.88), it performed less consistently under
340 challenging background conditions, with reductions in precision associated with increased false
341 detections. False positives were primarily associated with cloud boundaries, glare, and diffuse
342 atmospheric features rather than ground objects (Fig. 2). When results were aggregated across
343 recordings, consistent trade-offs between precision and recall were observed across strategies
344 and locations (Table 4).

345 Qualitative review of detection outputs identified recurring failure modes that explained
346 differences in precision and recall across strategies. These included false negatives associated
347 with low contrast, small apparent bird size, or partial obstruction; false positives caused by cloud
348 fragments, lens artifacts, or other non-bird motion; and temporal errors such as double-counting,
349 under-counting of clustered birds, and track fragmentation (Table 5).

350 Overall, appearance-based strategies exhibited consistent trade-offs between precision and recall
351 across locations.

352 *Effects of Motion-Based Discrimination*

353 Motion-based discrimination, applied following appearance-based filtering, altered detection
354 behavior relative to appearance-based filtering alone. Improvements in the balance between
355 precision and sensitivity were observed across both monitoring locations, as reflected in the
356 aggregated performance metrics (Table 4).
357 Across locations, inclusion of motion-based discrimination was associated with reductions in
358 false positives while largely preserving sensitivity to sustained bird flight. Improvements were
359 most pronounced at Location B, where background variability led to a higher frequency of
360 transient detections under appearance-based filtering, whereas gains at Location A were more
361 modest.

362 Fig. 2 showed that motion-based discrimination suppressed detections associated with diffuse
363 cloud structures and atmospheric artifacts while retaining detections corresponding to birds
364 exhibiting coherent flight trajectories.

365 Overall, motion-based discrimination reduced false detections while maintaining sensitivity to
366 sustained bird movement across locations.

367 *Temporal Confirmation and Detection Stability Using Sky-Based Adaptive Thresholding*

368 Temporal patterns of detected bird activity varied across monitoring locations when using the
369 sky-background filtering and adaptive threshold strategy. Daily detected bird counts,
370 summarized as the maximum number of concurrently detected birds per recording interval,
371 exhibited substantial variation over time, reflecting both changes in bird activity and background
372 conditions under field deployment.

373 Temporal confirmation through multi-frame tracking reduced short-term fluctuations in detected
374 counts and suppressed transient detections associated with atmospheric motion. Prior to temporal

375 filtering, detections derived from the sky-background and adaptive threshold pipeline showed
376 pronounced frame-to-frame variability, whereas post-confirmation counts displayed smoother
377 temporal trajectories and improved stability over consecutive recording intervals. These results
378 indicate that temporal confirmation improved the stability and interpretability of detection
379 outputs derived from the sky-background and adaptive threshold strategy.

380 *Location-Specific Variability in Detection Outputs*

381 Detection behavior differed across monitoring locations, reflecting variation in background
382 complexity and atmospheric conditions. Using the sky-background filtering and adaptive
383 thresholding strategy, Location A exhibited higher overall detected counts and greater variation
384 over time than Location B (Fig. 3). Temporal summaries based on the maximum number of
385 concurrently detected birds per recording interval showed location-specific variation in detected
386 bird activity over the monitoring period, with Location A showing higher overall counts and
387 Location B showing lower counts with intermittent spikes (Fig. 4). These findings indicate that
388 detection outputs varied by location, emphasizing the importance of context-specific
389 interpretation when applying computer vision-based monitoring systems.

390 **DISCUSSION**

391 This study evaluated a multi-stage computer vision pipeline for bird detection under field
392 conditions, examining the performance of appearance-based filtering, motion-based
393 discrimination, and temporal confirmation. Overall, the results demonstrate that detection
394 performance is strongly influenced by background complexity and that combining
395 complementary strategies is necessary to achieve robust performance across environments.
396 Annotation uncertainty represents an additional source of error not separately quantified in this
397 study. Manual counts were generated by two reviewers using continuous video playback, an

approach selected to match deployment input modality but subject to fatigue and subjective judgment at long viewing distances. The observed gap between automated and manual counts therefore reflects a combination of algorithmic limitations and annotation noise, particularly for small, distant, or partially occluded targets.

Appearance-based filtering. Appearance-based filtering strategies exhibited clear trade-offs between precision and recall across locations ($F1 = 0.69\text{--}0.88$). Sky-background filtering achieved higher precision (up to 0.83) by restricting detections to HSV-defined sky regions, but at the cost of reduced recall under low-contrast conditions, particularly when bird silhouettes overlapped with cloud structures. This limitation is consistent with prior work showing that sky segmentation performance degrades under real-world atmospheric variability, where changes in lighting, weather, and cloud cover alter pixel distributions and reduce segmentation reliability (Mihail et al., 2016; Place et al., 2019). Fixed-thresholding achieved relatively high precision (up to 0.92) under stable lighting but showed reduced consistency across recordings, indicating that a single global threshold cannot accommodate spatially heterogeneous illumination typical of outdoor environments. Such behavior is consistent with benchmarking studies demonstrating that fixed-threshold methods perform adequately under stable conditions but degrade under varying meteorological regimes (Hasenbalg et al., 2020). Adaptive thresholding improved sensitivity (recall up to 0.84) but introduced additional false positives associated with cloud-edge gradients and diffused atmospheric features. Collectively, these findings indicate that no single appearance-based strategy is universally optimal and that performance depends on environmental conditions and the relative importance of false positives versus missed detections.

Motion-based discrimination. The incorporation of motion-based discrimination substantially improved detection performance, increasing mean $F1$ from approximately 0.43 to 0.69 across

421 locations. This improvement was primarily driven by reductions in false positives while
422 maintaining sensitivity to sustained bird movement. Gains were most pronounced at Location B,
423 where background variability generated frequent transient detections under appearance-based
424 filtering alone. This improvement reflects the complementary information provided by motion
425 cues, which are less sensitive to illumination variability because they depend on temporal change
426 rather than absolute pixel intensity or color (Kim and Jung, 2017). As a result, detections that
427 satisfy appearance criteria in isolated frames but lack coherent trajectories are effectively
428 suppressed, while detections corresponding to sustained directional movement are retained. This
429 behavior is consistent with prior work demonstrating that combining motion and appearance cues
430 reduces false positives in dynamic outdoor environments (Ramirez et al., 2014; Wang and Yagi,
431 2012). Compared with appearance-based filtering alone, motion-based discrimination therefore
432 provides a critical mechanism for improving robustness under variable background and lighting
433 conditions.

434 **Temporal confirmation.** Temporal confirmation further improved detection stability by reducing
435 frame-to-frame variability and suppressing transient detections associated with atmospheric
436 motion. In our data, this was reflected in smoother temporal trajectories of detected bird counts
437 following multi-frame confirmation, particularly under conditions with high background
438 variability. This improvement reflects the use of temporal consistency, where true detections
439 persist across consecutive frames while spurious detections tend to occur only briefly. This
440 principle has been widely demonstrated in the literature, where multi-frame filtering reduces
441 false positives by requiring spatial and temporal coherence across frames (Hansen et al., 2005;
442 Khan et al., 2020). In the present study, enforcing persistence across frames resulted in smoother
443 temporal trajectories and improved interpretability of detected bird activity. However, temporal

444 filtering introduces an inherent trade-off, improving precision at the potential cost of missed
445 detections when objects appear briefly or exhibit irregular motion. Overall, these results indicate
446 that temporal confirmation provides a practical mechanism for stabilizing detection outputs in
447 dynamic outdoor environments.

448 ***Location-specific variability.*** Detection performance varied across monitoring locations despite
449 the use of identical pipeline parameters. Location A exhibited higher overall detection counts and
450 greater variation over time, whereas Location B showed lower but more stable detection patterns.
451 These differences were reflected in both the magnitude and temporal dynamics of detected bird
452 activity, indicating that background complexity and atmospheric conditions influenced detection
453 behavior independently of true bird presence. This pattern is consistent with prior work showing
454 that computer vision models often exhibit reduced performance when applied across different
455 environmental or geographic contexts due to domain shift in lighting, weather, and scene
456 composition (Hendrycks et al., 2021; Kalluri et al., 2023). A transient increase in detected counts
457 at Location A was associated with livestock movement within the camera field of view,
458 illustrating a limitation specific to agricultural environments where large non-target objects can
459 satisfy both appearance and motion criteria. Incorporating simple size-based filtering or non-
460 target exclusion rules could reduce such artifacts in future implementations.

461 ***Limitations***

462 The findings should be interpreted in the context of the study design. Among the limitations, the
463 evaluation was conducted at two monitoring locations under a defined set of environmental
464 conditions. However, these sites were selected to represent contrasting background complexity
465 and detection challenges, allowing assessment of pipeline performance across distinct field
466 scenarios. Data collection was limited to morning hours (8:00-11:00 am) during summer months,

467 restricting the range of illumination conditions encountered. The HSV-based sky classifier was
468 designed to handle four atmospheric states, but observed conditions were dominated by clear-sky
469 and light-overcast frames. Performance under dusk, dawn, low-angle sun, precipitation, and
470 winter overcast remains untested. The reported metrics therefore reflect a constrained lighting
471 regime, and generalization to round-the-clock or year-round deployment requires further
472 validation. The proposed pipeline uses rule-based thresholds and empirically tuned parameters,
473 which may require adjustment when deployed in new environments. On the other hand, this
474 design enables transparent control and practical implementation without the need for retraining.
475 Manual image-based annotations were used as a reference and, while not representing a complete
476 census of bird activity, provided a consistent basis for comparative evaluation across detection
477 strategies. Finally, non-target movement, such as livestock within the field of view, can influence
478 detection outputs when appearance and motion cues overlap, although these effects were
479 identifiable and did not affect the overall temporal patterns of bird activity. This study does not
480 include a direct deep learning benchmark. The pipeline is positioned as a complement to, rather
481 than replacement for, learned models. Deep approaches such as YOLO typically achieve higher
482 recall on sufficiently large labeled training sets but require GPU hardware, domain-matched
483 annotations, and retraining when camera position, species mix, or seasonal background changes.
484 The proposed pipeline targets the operational regime where these prerequisites are unavailable.
485 Direct head-to-head comparison under matched data conditions is identified as future work.

486 ***Practical implications***

487 The results demonstrate that combining region-of-interest filtering, motion-based discrimination,
488 and temporal confirmation provides a practical framework for quantifying bird activity under
489 field conditions. While detection accuracy varies with environmental context, the approach

captures temporal patterns that are directly relevant for monitoring bird pressure in agricultural systems. These findings support the use of lightweight, training-free computer vision pipelines as a practical option for on-farm surveillance where computational resources and annotated data are limited. However, successful deployment will depend not only on technical performance but also on how consistently the system is implemented and interpreted on farms. Practical use should therefore include site-specific calibration of motion-related parameters, because these settings govern the trade-off between transient false positives and missed brief events. In addition, integration of automated monitoring tools into farm management systems will likely require clear protocols and personnel training, as implementation quality can influence the effectiveness of dairy management interventions (Rodriguez et al., 2024a). Future work should evaluate user-adjustable parameter settings that allow site-specific calibration while preserving the transparency and computational simplicity of the pipeline.

CONCLUSION

Automated bird monitoring in agricultural environments presents persistent challenges due to heterogeneous sky backgrounds, atmospheric motion, and repeated detections across video frames. Rather than proposing a single optimized detection model, this study evaluates complementary computer vision design strategies applied sequentially within a practical detection pipeline. Using field-recorded video data from a commercial dairy facility, we assessed the effects of ROI filtering, appearance-based filtering, motion-based discrimination using optical flow, and temporal confirmation through multi-frame tracking. Performance was evaluated using manual image-based reference annotations to characterize detection behavior under variable environmental conditions.

512 Appearance-based strategies exhibited trade-offs between precision and sensitivity that varied by
513 background complexity, while motion-based discrimination reduced transient atmospheric
514 detections. Temporal confirmation improved detection stability and reduced double-counting
515 across frames. These findings highlight the value of sequential, multi-stage detection pipelines
516 for applied bird monitoring and provide practical guidance for deploying computer vision
517 systems under real-world agricultural conditions.

518

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589

591 Table 1: Implementation parameters for appearance-based detection pipelines by strategy.

Parameter	Sky-background filtering	Fixed threshold	Adaptive threshold
Resize factor	0.8	0.9	0.8
Gaussian kernel	5×5	3×3	5×5
Adaptive threshold block size	11	15	11
Constant offset	2	22	—
Min bird area	15	10	15
Max bird area	800	1000	800
Min aspect ratio	0.4	0.3	0.4
Max aspect ratio	3.0	4.0	3.0

Note: “—” indicates parameters not applicable to the given strategy.

Parameter values were selected empirically to optimize detection performance across recording conditions.

593

594 Table 2: Motion-based discrimination parameters by detection strategy

Parameter	Sky-background filtering	Fixed threshold	Adaptive threshold
Motion threshold	25	30	10
Min motion pixels	5	N/A	5
Min bird speed	3	N/A	3
Max bird speed	100	N/A	100

Note: Fixed-threshold uses simplified frame differencing based on mean motion intensity rather than dense optical flow; therefore, pixel-level motion parameters are not applicable.

595

596

597 Table 3. Temporal confirmation parameters for flight pattern recognition and tracking by
598 detection strategy

Parameter	Sky-background filtering	Fixed threshold	Adaptive threshold
Minimum flight duration	3 frames	5 frames	3 frames
Minimum movement distance	15 pixels	25 pixels	15 pixels
Maximum stationary frames	10	20	10
Maximum matching distance	80 pixels	100 pixels	80 pixels

599 Note: Minimum flight duration and movement distance thresholds were adjusted according to
600 frame processing rate and image scaling. For implementations processing every second frame,
601 shorter minimum durations were used to maintain temporal consistency. Movement distance
602 thresholds were scaled proportionally to the image resize factor.

603

604

605 Table 4. Aggregated detection performance of filtering strategies by location using manual
606 image-based reference counts.

Strategy	Location	Frames	TP	FP	FN	Precision	Recall	F1
Fixed threshold	A	27	481	41	93	0.92	0.84	0.88
Fixed threshold	B	24	64	8	32	0.89	0.67	0.77
Sky-background filtering	A	108	481	158	111	0.75	0.81	0.78
Sky-background filtering	B	16	64	13	40	0.83	0.62	0.71
Adaptive threshold	A	33	481	132	93	0.78	0.84	0.81
Adaptive threshold	B	58	64	26	32	0.71	0.67	0.69

607 Notes: Fixed-threshold processing used a constant threshold with standard background filtering;
608 sky-background filtering restricted detections to HSV-defined sky regions with cloud rejection;
609 and adaptive thresholding used a dynamically adjusted threshold with a context-dependent
610 region of interest (ROI).

611 Frames represent the number of manually annotated frames used for evaluation. Metrics were
612 calculated at the frame level using count-based agreement between automated detections and
613 manual reference counts. TP, FP, and FN represent summed count-based values across annotated
614 frames selected from the reviewed video segments.

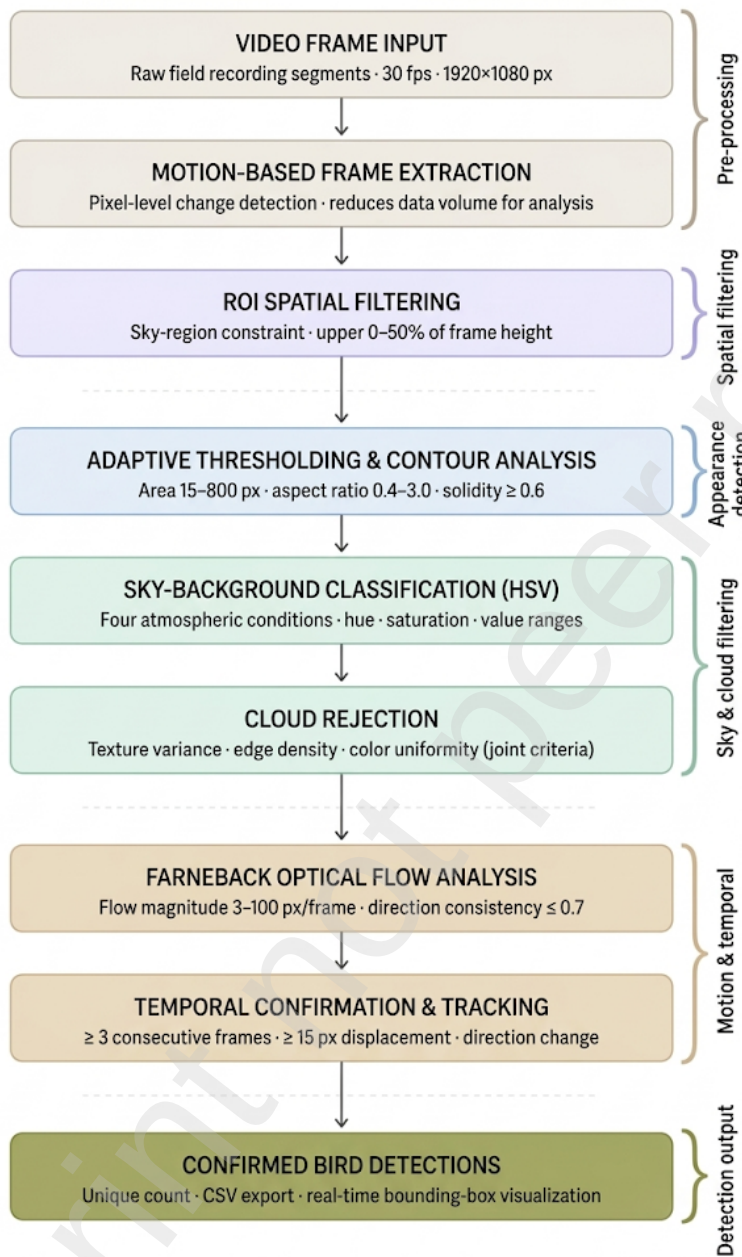
615 Abbreviations: TP = true positives; FP = false positives; FN = false negatives. Precision =
616 $TP/(TP + FP)$; Recall = $TP/(TP + FN)$; $F1 = 2 \times (Precision \times Recall)/(Precision + Recall)$.

618 Table 5. Qualitative summary of observed failure modes affecting bird-detection outputs under
 619 field conditions.

Failure mode	Likely cause	Visual evidence used for review
False negative: bird missed	Low contrast between bird and background; bird too small or distant; bird partially obscured by clouds or structures	Annotated bird visible in the raw frame but absent from confirmed detection output
False positive: non-bird object counted as bird	Cloud fragments, insects, lens artifacts, aircraft, moving shadows, or other objects meeting size, color, and motion criteria	Confirmed bird-detection box assigned to a non-bird object
Double-counting	Bird exits and re-enters the frame; two birds cross paths; tracker assigns a new identity to the same bird	Track interruption followed by a new detection ID for the same apparent bird
Under-counting	Multiple birds fly close together and are grouped into a single contour or track	One detection box includes two or more visually distinct birds
Track fragmentation	Bird temporarily disappears behind a cloud, structure, or low-contrast background and then reappears after the maximum stationary-frame threshold	Same apparent bird represented by two separate track segments
Motion confusion	Camera shake, moving vegetation, livestock, vehicles, or other non-target motion meets motion criteria	Motion-stage detections appear on non-bird moving objects

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622

623 Figure 1. Workflow of the training-free multi-stage computer vision pipeline for wild bird
 624 detection. Video frames were processed through region-of-interest filtering, appearance-based
 625 candidate detection, sky-background classification with cloud rejection, optical-flow-based

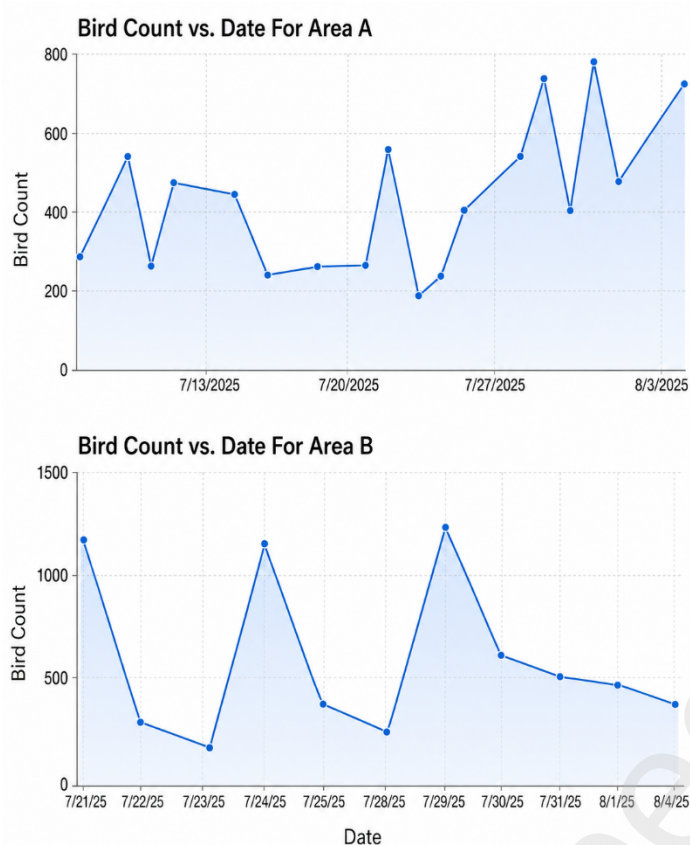
626 motion discrimination, and temporal confirmation. Thresholds and decision criteria are indicated
627 at each stage.



628
 629 Figure 2. Detection outputs at Location A using the sky-based adaptive-threshold pipeline. Left
 630 panels show raw video frames and right panels show detection overlays. Yellow boxes indicate
 631 appearance-based candidates, green boxes indicate candidates with bird-like motion, red boxes
 632 indicate confirmed flying-bird detections, and the purple line indicates the lower region-of-
 633 interest boundary.



635
 636 Figure 3. Detection outputs at Location B using the sky-based adaptive-threshold pipeline. Left
 637 panels show raw video frames and right panels show detection overlays. Yellow boxes indicate
 638 appearance-based candidates, green boxes indicate candidates with bird-like motion, red boxes
 639 indicate confirmed flying-bird detections, and the purple line indicates the lower region-of-
 640 interest boundary.



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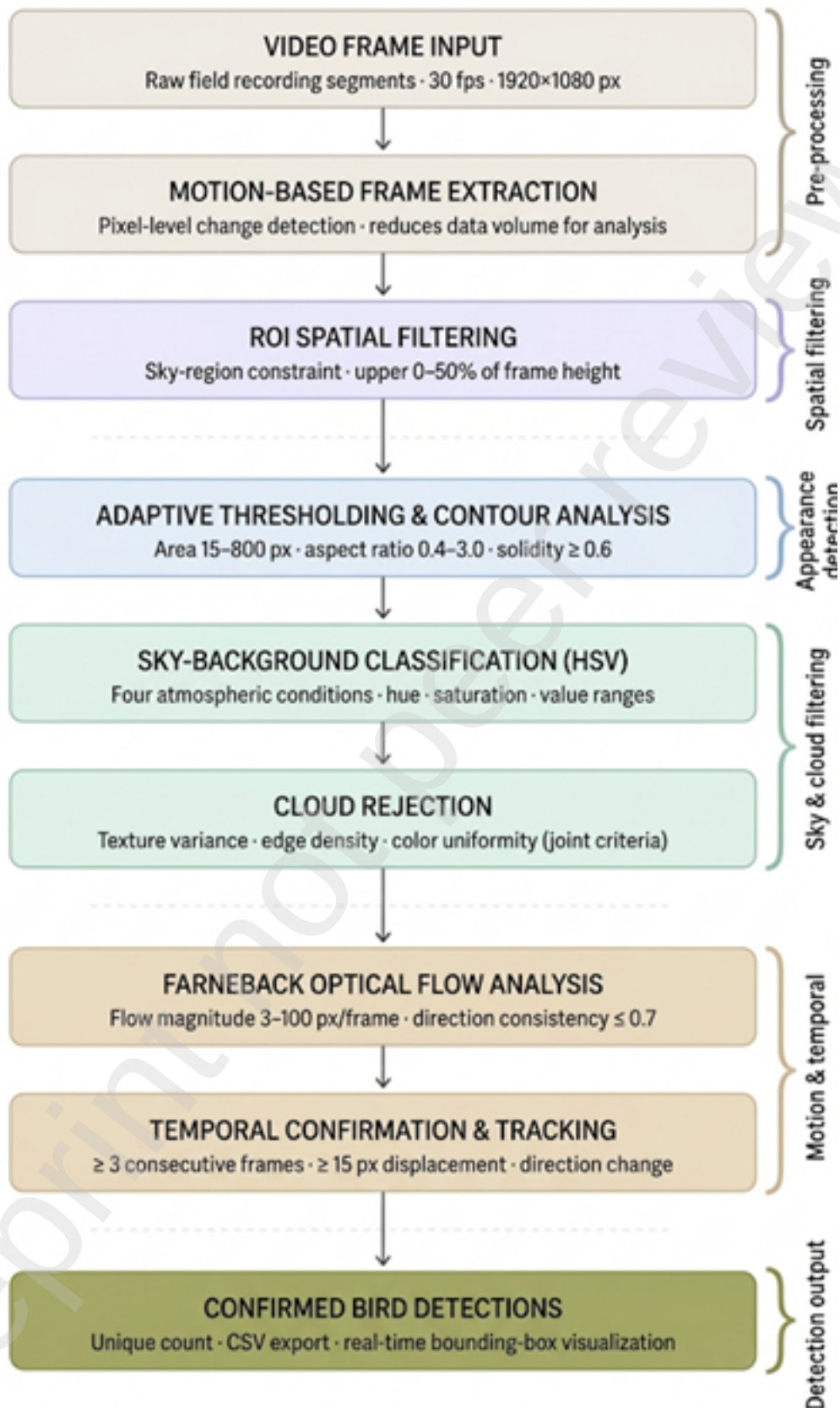
643 Figure 4. Temporal variation in detected bird counts at Locations A and B over the monitoring

644 period, based on the maximum number of concurrently detected birds per recording interval.

645 Location A exhibited higher overall counts, whereas Location B showed lower counts with

646 intermittent spikes.

647





Frame: 60/131
UNIQUE Flying Birds: 14
ROI Bottom: 54.1%
Detections: 282
Sky-filtered: 42
Moving: 14
Flying now: 14



Frame: 560/53948
UNIQUE Flying Birds: 26
ROI Bottom: 53.6%
Detections: 281
Sky-filtered: 42
Moving: 10
Flying now: 9





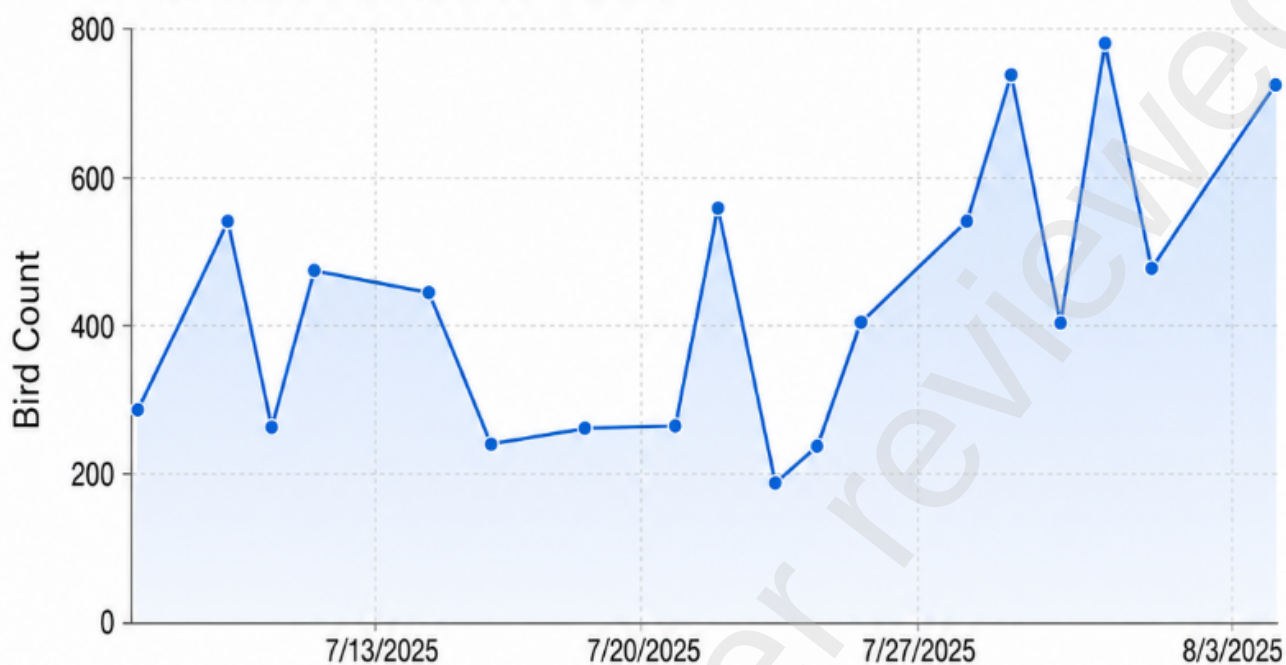
Frame: 110/233
UNIQUE Flying Birds: 43
ROI Bottom: 65.1%
Detections: 351
Sky-filtered: 92
Moving: 31
Flying now: 28



Frame: 190/233
UNIQUE Flying Birds: 67
ROI Bottom: 80.0%
Detections: 326
Sky-filtered: 91
Moving: 19
Flying now: 19



Bird Count vs. Date For Area A



Bird Count vs. Date For Area B

